

ERAP1 × axial spondyloarthritis

AXIS TARGET DECISION BRIEF | Executive decision | v1.0

Assessment: 5 October 2026. Frozen evidence snapshot through 5 October 2026; selected corpus, not a comprehensive literature cutoff. Primary disease evidence concerns ankylosing spondylitis (AS); broader axSpA applicability is unassessed.

CURRENT AXIS POSITION

Scientific position: **PARTIALLY SUPPORTED** for investigating context-specific ERAP1 modulation. Therapeutic benefit from ERAP1 inhibition is not established in the current AXIS evidence scope.

Strongest evidence: AS genetic association and HLA-context interaction, plus source-reported peptide-processing and cellular perturbation effects.

Strongest concern: cellular phenotypes lack established direct engagement; phenotype direction differs between cell systems.

Critical uncertainty: Do the phenotype-producing compounds directly engage ERAP1 in cells at those exposures?

Recommended next experiment: Matched-genotype chemical-genetic experiment with an engagement readout.

Current decision: **RESOLVE CRITICAL UNCERTAINTY FIRST.**

Why a scientific decision-maker should care

ERAP1 has a plausible evidence chain to AS biology, but attributing a compound-induced phenotype to cellular ERAP1 is the unresolved step that determines whether to invest in that chemical strategy.

Independent scientific review pending.

Evidence chain & chemical opportunity



*Experimental structure supported: therapeutic relevance and druggability unestablished.

Evidence layer	AXIS position	Key evidence	Main limitation	Decision relevance
Genetics	SUPPORTED	AS association; HLA interaction	Association ≠ therapeutic direction	Biological rationale
Target-disease	PARTIAL	Genetics + cellular mechanism	Broader axSpA unassessed	Scope program
Mechanism	SUPPORTED	Peptide trimming / repertoire	Allotype and substrate context	Avoid global activity rule
Perturbation	MIXED	Genetic + chemical effects	Opposite FHC directions	Test attribution
Structure	SUPPORTED*	3QNF open state	Missing coordinates; no drug ligand	Bound structure use
Chemical matter	PARTIAL	Maben 1-3	Small set; identity caveats	Reference chemistry
Biochemical	SUPPORTED	Context-specific activity	IC50, AC50, Ki differ	Fix assay context
Cellular	PARTIAL	FHC / presentation / Th17	Systems are not equivalent	Separate phenotype
Engagement	INSUFFICIENT	Not assessed directly	Decision-blocking gap	Measure in cells
Selectivity	INSUFFICIENT	Counter-screen bounds	Comparability unresolved	No ratio claim
Translation	INSUFFICIENT	AS T-cell coculture	No established disease rescue	Further validation
Clinical	NOT ASSESSED	No efficacy indexed	Selected corpus only	No efficacy conclusion

AXIS editorial layer states are categorical summaries, not numeric maturity scores. *Structure existence does not establish druggability. FHC = free heavy chain.

CHEMISTRY: three experimental Maben compounds; three computational exploration candidates (bestatin, captopril, vorinostat). Candidate identity is externally verified, but ERAP1 pharmacology is unestablished. Docking was not executed.

MODEL NOT BUILT: 15 measurements, seven assays, nine distinct assay/endpoint datasets; all nine refused predictive modelling. Identity, prediction and experiment stay separate.

The next decision

THE DECISION BOTTLENECK

Biochemical activity + cellular phenotype



On-target / off-target / indirect / context-dependent explanations



Direct cellular engagement remains unestablished



Matched-genotype engagement + phenotype experiment



Strengthen, weaken, revise mechanism, or repeat after technical validation

WHY THIS EXPERIMENT NEXT?

- It separates three pairs of viable explanations using engagement and genetic dependency together.
- It can weaken the preferred on-target explanation, rather than merely gather supporting observations.
- It connects distinct interpretable outcomes to different scientific actions.
- It uses a target-proximal endpoint with genetic and viability controls; engagement alone would not establish phenotype dependency.

Outcome	Interpretation and decision consequence
A: engagement + dependency + rescue	Strengthens ERAP1-dependent action for this compound and context. Selectivity, indirect mechanism and broader translation remain open.
B: phenotype persists without ERAP1	Weakens ERAP1 attribution for this compound; pursue another chemotype or explanation. Not universal target failure.
C: invalid assay/model or toxicity	Non-interpretable. Keep uncertainty open; validate controls and repeat.

WHAT WOULD CHANGE OUR MIND?

Strengthen: engagement aligned with active exposure, genetic dependency and rescue. Weaken: valid engagement-negative result or ERAP1-independent phenotype. Invalidate the local dependency interpretation: reproducible phenotype persistence in a validated ERAP1-null model. Leave unresolved: failed controls, toxicity, or positive engagement without dependency.

Prerequisites: validated chemical identity, engagement assay and null/rescue model; a reference engaged ligand and inactive analogue are not established in this corpus. Cost, timing and local feasibility are unknown.

Selected references: Evans 2011 (doi:10.1038/ng.873); Chen 2014 (doi:10.1002/art.38249); Chen 2016 (doi:10.1136/annrheumdis-2014-206996); Tran 2016 (doi:10.1016/j.molimm.2016.04.002); Maben (doi:10.1021/acs.jmedchem.9b00293). Independent scientific review pending.